Experimental Protocol: Measuring Intrinsic Clock-Induced Dephasing (\Gamma_{jk}) via a Superconducting Qubit-Resonator Quantum Clock

To test the finite-clock corrections of the **Hilbert-Space Quadrature Conjecture**, we must construct an experimental realization of a bipartite quantum clock-system composite \mathcal{H}_U = \mathcal{H}_C \otimes \mathcal{H}_S where the clock's energy spread \Delta E_C can be continuously tuned, and the resulting non-unitary dephasing rate \Gamma_{jk} = \frac{(E_j - E_k)^2}{2\hbar \Delta E_C} can be isolated from standard environmental noise channels.
Here is a concrete superconducting circuit QED implementation.

I. Experimental Architecture

The physical system consists of a transmon qubit acting as the quantum system (\mathcal{H}_S) coupled dispersively to a high-coherence microwave coplanar waveguide (CPW) resonator acting as the quantum clock (\mathcal{H}_C).

1. **The Clock Sector (\mathcal{H}_C):** A high-Q superconducting LC resonator with resonance frequency \omega_c / 2\pi \approx 6.0\text{ GHz} and characteristic impedance Z_0. The photon number operator is \widehat{N} = a^\dagger a, with an adjustable mean photon number (energy spread) controlled by a calibrated coherent drive tone applied prior to the interaction window. The effective clock energy spread is \Delta E_C = \hbar \omega_c \sqrt{\langle (a^\dagger a - \langle a^\dagger a \rangle)^2 \rangle}.
2. **The System Sector (\mathcal{H}_S):** A frequency-tunable transmon qubit with ground state \vert{}0\rangle and excited state \vert{}1\rangle, having transition frequency \omega_s tuned via magnetic flux threading its SQUID loop.
3. **The Interaction Hamiltonian:** Dispersive coupling between the clock resonator and the transmon qubit is given by: \widehat{H}_{\text{int}} = \hbar \chi \, a^\dagger a \, \vert{}1\rangle\langle 1\vert{}, where \chi / 2\pi \approx -1.0\text{ MHz} is the dispersive shift. In the interaction picture, this realizes the discrete Page-Wootters constraint where the effective time elapsed for the qubit depends directly on the photon-number phase accumulation of the resonator clock.

II. Protocol: State-Generated Quadrature Evolution & Readout

To measure the clock-induced dephasing rate \Gamma_{01} as a function of clock uncertainty \Delta E_C:

1. **State Preparation:**
    - Initialize the transmon qubit in the superposition state \vert{}\psi_0\rangle = \frac{1}{\sqrt{2}}(\vert{}0\rangle + \vert{}1\rangle) using a \frac{\pi}{2} microwave pulse.
    - Initialize the resonator clock in a coherent state \vert{}\alpha\rangle with mean photon number \bar{n} = \vert{}\alpha\vert{}^2 (where \bar{n} is tuned across values from 10 to 10^4). The clock energy spread is \Delta E_C = \hbar \omega_c \sqrt{\bar{n}}.

2. **Relational Evolution Window:**
    - Turn off external drives and allow the bipartite system to evolve freely under the joint Hamiltonian for an interaction time t. Because the resonator photon number n fluctuates by \Delta n = \sqrt{\bar{n}}, each photon number component n imparts a distinct conditional phase shift \phi_n = n \chi t onto the qubit.
3. **Quadrature Readout (Phase-Space Tomography):**
    - Apply a heterodyne measurement on the resonator (projecting onto the covariant phase POVM E(d\theta)) to register the clock reading \theta, while simultaneously performing state tomography on the transmon qubit to extract the off-diagonal coherence element \rho_{01}(\theta, t).

III. Isolating the Intrinsic Rate from Environmental Decoherence

The measured decoherence rate \gamma_{\text{meas}} will contain both standard environmental decoherence (\gamma_{\text{env}}, due to dielectric loss and photon leakage \kappa) and the conjectured intrinsic clock-induced dephasing (\Gamma_{01}).
To isolate \Gamma_{01} = \frac{(E_1 - E_0)^2}{2\hbar \Delta E_C} = \frac{(\hbar \chi)^2}{2\hbar (\hbar \omega_c \sqrt{\bar{n}})}:

1. **Scaling Analysis:**
    - According to standard environmental decoherence models (e.g., thermal photon shot noise), dephasing scales linearly or independently with mean photon number in specific regimes.
    - Under the Quadrature Conjecture's finite-clock master equation (Theorem 2), the intrinsic dephasing rate scales inversely with the clock energy spread: \Gamma_{01}(\bar{n}) = \frac{\chi^2}{2 \omega_c \sqrt{\bar{n}}}.
2. **Differential Measurement:**
    - By sweeping the clock energy spread \Delta E_C (via \bar{n}) while holding the environmental temperature and cavity decay rate \kappa constant, the 1/\sqrt{\bar{n}} dependence isolates the state-generated quadrature dephasing signature from background environmental noise.